

DARCIEA ARULNESAN

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AI Engineer who builds agentic LLM systems end-to-end, from stakeholder requirements to production deployment. MEng (First Class), Imperial College London, with a research background in human-robot interaction.

WORK EXPERIENCE

Associate AI Innovation Engineer, Dojo

September 2025 - March 2026

- Architected an agentic AI system for merchant risk assessment, scoping requirements with product stakeholders and orchestrating LLM chains with external API integration and structured output parsing to deliver end-to-end automation.
- Built a RAG-powered knowledge assistant on Vertex AI using LangChain and embedding-based retrieval, enabling CS advisors to query internal documentation via natural language, replacing manual document searches.
- Deployed AI services to GCP using Terraform for infrastructure-as-code and GitHub Actions for CI/CD, building production-grade pipelines within an Akka-based microservices architecture.

Research Intern, PAIR Lab, Imperial College London

June - August 2025

- Designed and ran a human-robot interaction study in the Personal Assistive and Intelligent Robotics (PAIR) Lab, investigating how robot authority roles (peer vs. leader) influence human decision-making in collaborative rescue scenarios.
- Built a mission-style experimental interface in Python (Tkinter) and programmed Nao and Misty robots via their Python SDKs to deliver autonomous verbal and gestural behaviours for controlled, repeatable trials.

EDUCATION

Imperial College London, MEng Mechanical Engineering

2019 - 2024

- First Class Honours
- Relevant modules: Machine Learning, Advanced Control, Robotics, Human Neuromechanical Control and Learning

Wimbledon High School, London

2012 - 2019

- A Levels: Maths (A*), Further Maths (A), Physics (A), Chemistry (A)

TECHNICAL PROJECTS

MNIST Digit Classifier

April - May 2025

- Trained a CNN digit classifier in PyTorch (98.4% accuracy) and deployed a full-stack Streamlit app with PostgreSQL logging, containerised end-to-end with Docker on a self-managed server.

MEng Individual Project — A Wearable Sensor to Measure Fetal Movements

2023 - 2024

- Clinically validated a novel multi-modal wearable fetal movement monitor developed by the Imperial College Biomechatronics lab, enabling accurate at-home monitoring.
- Trained a transformer-based binary classifier in TensorFlow on multi-modal sensor data (50+ participants), improving fetal movement detection accuracy, evaluated against clinical ultrasound ground truth.

SKILLS

Programming	Python, SQL, Java, C, MATLAB
ML / AI	LangChain, PyTorch, TensorFlow, HuggingFace, Vertex AI, RAG, Embeddings
Tools & Infra	Git, Docker, Terraform, GCP, Akka, Linux, Jira
Spoken Languages	English, Spanish, Tamil